

M21_20260502

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Load libraries

Set emmeans options

```
emm_options(
  lmer.df = "kenward-roger",
  pbkrtest.limit = 10000,
  disable.pbkrtest = FALSE
)
```

Set ggplot parameters

```
# Color palette
my_palette_1 <- c(
  "#E31A1C", # red
  "#1F78B4" # blue
)

my_palette_2 <- c(
  "#0072B2", # dark blue
  "#E69F00" # orange/gold
)
my_linetypes_1 <- c("dashed", "solid")

# Theme
my_theme <- theme_classic(base_size = 10) +
  theme(
    legend.position = "bottom",
    legend.direction = "horizontal",
    legend.box = "horizontal"
  )

# Common guides
my_guides <- guides(
  colour = guide_legend(nrow = 1),
  fill = guide_legend(nrow = 1)
)
```

Define 2-way interaction table function

1 Load Files

```
# Load data: rt words
rt_w <- read_csv("rt_words_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    Item = factor(target),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: rt nonwords
rt_nw <- read_csv("rt_nwords_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity,
      levels = c("SIMP", "COMP"),
      labels = c("Simple", "Complex")),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: erp_fs words
erp_fs_w <- read_csv("erp_fs_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
```

```

        levels = c("Small", "Large")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_fs nonwords
erp_fs_nw <- read_csv("erp_fs_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf words
erp_bf_w <- read_csv("erp_bf_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf nonwords
erp_bf_nw <- read_csv("erp_bf_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

range(erp_fs_nw$LP)

```

```

|| [1] -2.674523  3.111265
range(erp_fs_nw$OS)

```

```

|| [1] -1.975820  1.926921

```

2 RESPONSE TIME

2.1 RESPONSE TIME WORD DATA

2.1.1 Model

```

library(lmerTest)

m_rt_words <- lmer(
  logRT ~ zLogBF + zLogFS + OS + LP +
    (1 | SubjID) +
    (1 | Item),
  data = rt_w,
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_rt_words)

```

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ zLogBF + zLogFS + OS + LP + (1 | SubjID) + (1 | Item)
|| Data: rt_w
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -3536.3
||
|| Scaled residuals:

```

```

||      Min      1Q  Median      3Q      Max
|| -9.8884 -0.6626 -0.0643  0.5951  5.1184
||
|| Random effects:
|| Groups   Name                Variance Std.Dev.
|| Item     (Intercept) 0.001807 0.04251
|| SubjID    (Intercept) 0.017718 0.13311
|| Residual                    0.029621 0.17211
|| Number of obs: 5847, groups: Item, 100; SubjID, 67
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)   6.390069   0.016965   72.434536 376.655 < 2e-16 ***
|| zLogBF        -0.017184   0.004696  100.974079  -3.659 0.000404 ***
|| zLogFS        -0.012858   0.004865   93.591020  -2.643 0.009629 **
|| OS            -0.034712   0.018734   63.842814  -1.853 0.068518 .
|| LP            0.011591   0.012386   63.973484   0.936 0.352893
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) zLogBF zLogFS OS
|| zLogBF      0.009
|| zLogFS      0.002 -0.105
|| OS          0.012 -0.001  0.001
|| LP          0.008  0.000  0.001  0.008

```

Effect sizes for individual terms
`eta_squared(m_rt_words, partial = TRUE)`

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) |      95% CI
|| -----|-----|-----|
|| zLogBF    |      0.12 | [0.04, 1.00]
|| zLogFS    |      0.07 | [0.01, 1.00]
|| OS        |      0.05 | [0.00, 1.00]
|| LP        |      0.01 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

Model R²
`r2_nakagawa(m_rt_words)`

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.417
|| Marginal R2: 0.033

```

2.2 RESPONSE TIME NONWORD DATA

2.2.1 Model

```

contrasts(rt_nw$complexity) <- c(-0.5, 0.5)

m_rt_nw <- lmer(
  logRT ~ complexity + zLogBF + zLogFS + OS + LP +
    (1 + complexity | SubjID) +
    (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)
m_rt_nw

|| Linear mixed model fit by REML ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity | SubjID) + (1 | ItemID)
|| Data: rt_nw
|| REML criterion at convergence: -4165.782
|| Random effects:
|| Groups   Name                Std.Dev. Corr
|| ItemID    (Intercept) 0.03662
|| SubjID     (Intercept) 0.13143
||            complexity1 0.02365 -0.17
|| Residual                    0.14641
|| Number of obs: 4607, groups: ItemID, 100; SubjID, 67
|| Fixed Effects:
|| (Intercept) complexity1      zLogBF      zLogFS      OS      LP
||      6.554486      0.048807      0.015469     -0.006359     -0.041063      0.003398

```

`summary(m_rt_nw)`

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity | SubjID) + (1 | ItemID)
|| Data: rt_nw
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -4165.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -4.2605 -0.6625 -0.0227  0.6633  3.8775
||
|| Random effects:
||  Groups      Name                Variance Std.Dev. Corr
||  ItemID      (Intercept)  0.0013408  0.03662
||  SubjID      (Intercept)  0.0172747  0.13143
||              complexity1  0.0005594  0.02365  -0.17
||  Residual                    0.0214360  0.14641
|| Number of obs: 4607, groups: ItemID, 100; SubjID, 67
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)  6.554486   0.016639  68.832580 393.923 < 2e-16 ***
|| complexity1  0.048807   0.005347  62.675281   9.128 4.07e-13 ***
|| zLogBF       0.015469   0.005080  98.214341   3.045 0.00299 **
|| zLogFS      -0.006359   0.005137  96.636227  -1.238 0.21880
|| OS          -0.041063   0.018454  62.212180  -2.225 0.02970 *
|| LP           0.003398   0.012228  62.906744   0.278 0.78204
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 zLogBF zLogFS OS
|| complexity1 -0.073
|| zLogBF      -0.004 -0.005
|| zLogFS      -0.004  0.011 -0.576
|| OS           0.011 -0.007 -0.001 -0.001
|| LP           0.003 -0.015 -0.002  0.000  0.009
||
|| # Effect sizes for individual terms
|| eta_squared(m_rt_nw, partial = TRUE)
||
|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| complexity | 0.57 | [0.44, 1.00]
|| zLogBF     | 0.09 | [0.02, 1.00]
|| zLogFS     | 0.02 | [0.00, 1.00]
|| OS         | 0.07 | [0.00, 1.00]
|| LP         | 1.23e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
|| r2_nakagawa(m_rt_nw)
||
|| # R2 for Mixed Models
||
|| Conditional R2: 0.494
|| Marginal R2: 0.050
||
|| # Means
|| # get means on log scale
|| emm_logrt <- emmeans(m_rt_nw, ~ complexity)
|| emm_summary <- summary(emm_logrt)
||
|| # back transform manually
|| emm_summary$mean_ms <- exp(emm_summary$emmean)
|| emm_summary$lower_ms <- exp(emm_summary$lower.CL)
|| emm_summary$upper_ms <- exp(emm_summary$upper.CL)
|| emm_summary
||
|| complexity emmean SE df lower.CL upper.CL mean_ms lower_ms upper_ms
|| Simple 6.529 0.0172 70.03 6.495 6.563 684.6 661.5 708.5
|| Complex 6.578 0.0168 70.15 6.544 6.611 718.9 695.2 743.4
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
||
|| diff_ms <- exp(emm_summary$emmean[emm_summary$complexity == "Complex"]) -
|| exp(emm_summary$emmean[emm_summary$complexity == "Simple"])
|| diff_ms

```

```
|| [1] 34.24345
```

3 N250

3.1 N250 WORD DATA: Base Frequency

3.1.1 Model

```
contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N250_bf <- lmer(
  value ~ base_frequency * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_bf)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 18999.1
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -7.4496 -0.3820 -0.0038  0.3905 14.0821
||
|| Random effects:
|| Groups           Name             Variance Std.Dev.
|| SubjID:chlabel    (Intercept) 19.723   4.441
|| SubjID            (Intercept) 12.044   3.470
|| Residual                          8.095   2.845
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -1.5059    0.4650   56.9998  -3.239  0.00201 **
|| base_frequency1 -0.3183    0.1001 1617.9999  -3.181  0.00150 **
|| OS              0.5341    0.5200   56.9998   1.027  0.30874
|| LP             -0.5356    0.3781   56.9998  -1.417  0.16203
|| base_frequency1:OS -0.4549    0.1120 1617.9999  -4.062 5.09e-05 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) bs_fr1 OS      LP
|| bas_frqn1  0.000
|| OS         -0.046 0.000
|| LP          0.038 0.000 0.021
|| bs_frqn1:OS 0.000 -0.047 0.000 0.000
||
|| # Effect sizes for individual terms
|| eta_squared(m_w_N250_bf, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| base_frequency | 6.21e-03 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]
|| LP | 0.03 | [0.00, 1.00]
|| base_frequency:OS | 0.01 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
|| r2_nakagawa(m_w_N250_bf)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.801
|| Marginal R2: 0.018
```

3.1.2 Means and contrasts

```
emmeans(m_w_N250_bf, ~ base_frequency)

|| base_frequency emmean SE df lower.CL upper.CL
|| Low -1.29 0.467 58.3 -2.22 -0.355
|| High -1.63 0.467 58.3 -2.56 -0.692
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

emm_bf_OS <- emmeans(m_w_N250_bf,
  ~ base_frequency | OS,
  at = list(OS = c(-1, 0, 1)))

emm_part <- summary(emm_bf_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  ))

|| OS Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| 1 -1 Contrast: Low - High -0.14 0.15 1618.0 NA NA -0.89 0.374
|| 2 -1 EMM: High -1.95 0.72 58.3 -3.38 -0.51 NA NA
|| 3 -1 EMM: Low -2.08 0.72 58.3 -3.52 -0.65 NA NA
|| 4 0 Contrast: Low - High 0.32 0.10 1618.0 NA NA 3.18 0.001
|| 5 0 EMM: High -1.64 0.47 58.3 -2.57 -0.70 NA NA
|| 6 0 EMM: Low -1.32 0.47 58.3 -2.26 -0.39 NA NA
|| 7 1 Contrast: Low - High 0.77 0.15 1618.0 NA NA 5.27 0.000
|| 8 1 EMM: High -1.33 0.68 58.3 -2.70 0.04 NA NA
|| 9 1 EMM: Low -0.56 0.68 58.3 -1.93 0.81 NA NA

# kable(emm_contrast_table, digits = 3)
```

3.1.3 Plots

3.2 N250 WORD DATA: Family Size

3.2.1 Model

```
contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N250_fs <- lmer(
  value ~ family_size * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_fs)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15026.8
||
|| Scaled residuals:
|| Min 1Q Median 3Q Max
|| -8.4669 -0.4045 -0.0218 0.3824 12.9276
||
```

```

|| Random effects:
|| Groups      Name      Variance Std.Dev.
|| SubjID:chlabel (Intercept) 4.625    2.151
|| SubjID      (Intercept) 2.918    1.708
|| Residual                2.754    1.659
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)   -0.75979    0.22921   57.00000   -3.315   0.0016 **
|| family_size1    0.32357    0.05837  1617.99999    5.543 3.46e-08 ***
|| OS             0.26085    0.25635   57.00000    1.018   0.3132
|| LP            -0.27860    0.18636   57.00000   -1.495   0.1404
|| family_size1:OS  0.11120    0.06532  1617.99999    1.702   0.0889 .
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) fmly_1 OS      LP
|| family_size1  0.000
|| OS           -0.046  0.000
|| LP            0.038  0.000  0.021
|| fmly_sz1:OS   0.000 -0.047  0.000  0.000

```

Effect sizes for individual terms
`eta_squared(m_w_N250_fs, partial = TRUE)`

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----|-----|-----|
|| family_size    |      0.02 | [0.01, 1.00]
|| OS             |      0.02 | [0.00, 1.00]
|| LP             |      0.04 | [0.00, 1.00]
|| family_size:OS | 1.79e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

Model R²
`r2_nakagawa(m_w_N250_fs)`

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.738
|| Marginal R2: 0.019

```

3.2.2 Means and Contrasts

```

# family size
emmeans(m_w_N250_fs, ~ family_size)

```

```

|| family_size emmean    SE  df lower.CL upper.CL
|| Small      -0.900 0.231 58.9    -1.36   -0.438
|| Large      -0.571 0.231 58.9    -1.03   -0.110
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

```

3.2.3 Plots

3.3 N250 WORD DATA: Combined Plots

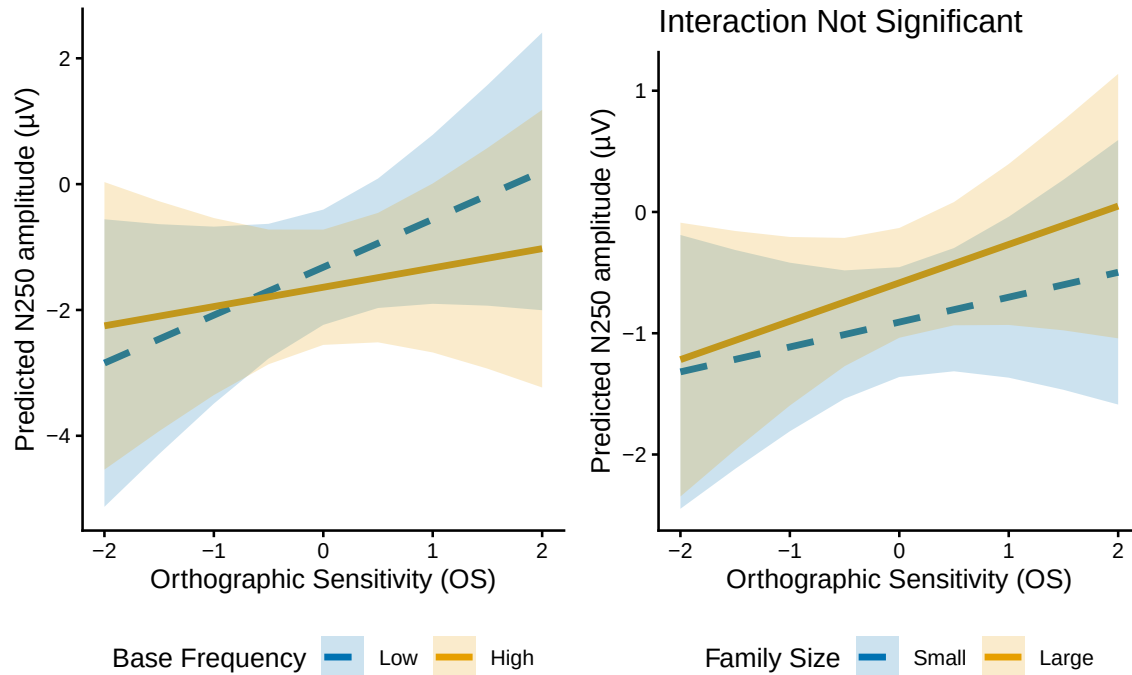
```

row1 <- plot_grid(BFxOS_n250,FSxOS_n250,ncol = 2)

plot_grid(
  ggdraw() + draw_label("N250 Word Data", fontface = "bold"),
  row1,
  ncol = 1,
  rel_heights = c(0.08, 1)
)

```


N250 Word Data



3.4 N250 NONWORD DATA: Base Frequency as covariate (NOT USED IN WRITE-UP)

3.4.1 Model

```
contrasts(erp_bf_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_bf_nw$base_frequency) <- c(-0.5, 0.5)

m_nw_N250_bf <- lmer(
  value ~ complexity * OS + complexity * LP + base_frequency +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N250_bf)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * OS + complexity * LP + base_frequency + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_nw, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 41527.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -12.5220  -0.4882  -0.0036   0.4647  24.5889
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept) 10.462    3.235
|| SubjID       (Intercept)  9.777    3.127
|| Residual                    27.474    5.242
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)   -1.4173    0.4175   57.0000   -3.395  0.00126 **
|| complexity1   -0.2243    0.1305  4856.0003   -1.719  0.08566 .
|| OS             0.5743    0.4669   57.0000    1.230  0.22379
|| LP            -0.3205    0.3394   57.0000   -0.944  0.34909
|| base_frequency1 -0.1639    0.1302  4856.0003   -1.258  0.20835
|| complexity1:OS  0.6090    0.1459  4856.0003    4.173  3.05e-05 ***
|| complexity1:LP  0.7230    0.1061  4856.0003    6.816  1.05e-11 ***
```

```

|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 OS      LP      bs_fri cm1:OS
|| complexity1  0.000
|| OS          -0.046  0.000
|| LP          0.038  0.000  0.021
|| bas_frqn1  0.000  0.000  0.000  0.000
|| cmplxty1:OS 0.000 -0.046  0.000  0.000  0.000
|| cmplxty1:LP 0.000  0.038  0.000  0.000  0.000  0.021
||
|| # Effect sizes for individual terms
|| eta_squared(m_nw_N250_bf, partial = TRUE)
||
|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----
|| complexity      |      6.08e-04 | [0.00, 1.00]
|| OS              |      0.03      | [0.00, 1.00]
|| LP              |      0.02      | [0.00, 1.00]
|| base_frequency  |      3.26e-04 | [0.00, 1.00]
|| complexity:OS   |      3.57e-03 | [0.00, 1.00]
|| complexity:LP   |      9.48e-03 | [0.01, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
|| r2_nakagawa(m_nw_N250_bf)
||
|| # R2 for Mixed Models
||
|| Conditional R2: 0.433
|| Marginal R2: 0.015

```

3.4.2 Means and Contrasts

```

## complexity
emmmeans(m_nw_N250_bf, ~ complexity, )

|| complexity emmean   SE   df lower.CL upper.CL
|| Simple      -1.26 0.422 59.8   -2.10   -0.417
|| Complex     -1.49 0.422 59.8   -2.34   -0.651
||
|| Results are averaged over the levels of: base_frequency
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
## complexity x LP
emm_bf_LP <- emmeans(m_nw_N250_bf,
  ~ complexity | LP,
  at = list(LP = c(-2, -1, 0, 1, 2)))

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table_bf_LP_n250 <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  ))

|| LP      Effect Estimate   SE      df lower.CL upper.CL t.ratio p.value
|| 1 -2 Contrast: Simple - Complex  1.64 0.24 4856.0      NA      NA    6.72 0.000
|| 2 -2      EMM: Complex   -1.57 0.79  59.8    -3.16   0.01      NA      NA
|| 3 -2      EMM: Simple    0.07 0.79  59.8    -1.51   1.65      NA      NA
|| 4 -1 Contrast: Simple - Complex  0.92 0.16 4856.0      NA      NA    5.59 0.000

```

```

|| 5 -1 EMM: Complex -1.53 0.53 59.8 -2.60 -0.47 NA NA
|| 6 -1 EMM: Simple -0.61 0.53 59.8 -1.68 0.46 NA NA
|| 7 0 Contrast: Simple - Complex 0.20 0.13 4856.0 NA NA 1.53 0.127
|| 8 0 EMM: Complex -1.49 0.42 59.8 -2.34 -0.65 NA NA
|| 9 0 EMM: Simple -1.29 0.42 59.8 -2.14 -0.45 NA NA
|| 10 1 Contrast: Simple - Complex -0.52 0.17 4856.0 NA NA -3.06 0.002
|| 11 1 EMM: Complex -1.45 0.55 59.8 -2.56 -0.34 NA NA
|| 12 1 EMM: Simple -1.98 0.55 59.8 -3.09 -0.87 NA NA
|| 13 2 Contrast: Simple - Complex -1.25 0.25 4856.0 NA NA -4.92 0.000
|| 14 2 EMM: Complex -1.41 0.82 59.8 -3.05 0.23 NA NA
|| 15 2 EMM: Simple -2.66 0.82 59.8 -4.30 -1.02 NA NA

```

```

## complexity x OS
emm_bf_OS <- emmeans(m_nw_N250_bf,
  ~ complexity | OS,
  at = list(OS = c(-1, 0, 1)))

emm_part <- summary(emm_bf_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  ))

```

```

|| OS Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| 1 -1 Contrast: Simple - Complex 0.87 0.20 4856.0 NA NA 4.34 0.000
|| 2 -1 EMM: Complex -2.41 0.65 59.8 -3.71 -1.11 NA NA
|| 3 -1 EMM: Simple -1.54 0.65 59.8 -2.84 -0.25 NA NA
|| 4 0 Contrast: Simple - Complex 0.26 0.13 4856.0 NA NA 1.99 0.047
|| 5 0 EMM: Complex -1.53 0.42 59.8 -2.38 -0.69 NA NA
|| 6 0 EMM: Simple -1.27 0.42 59.8 -2.12 -0.43 NA NA
|| 7 1 Contrast: Simple - Complex -0.35 0.19 4856.0 NA NA -1.83 0.067
|| 8 1 EMM: Complex -0.65 0.62 59.8 -1.89 0.59 NA NA
|| 9 1 EMM: Simple -1.00 0.62 59.8 -2.24 0.24 NA NA

```

3.4.3 Plots

3.5 N250 NONWORD DATA: Family Size as covariate

3.5.1 Model

```

contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)

m_nw_N250_fs <- lmer(
  value ~ complexity * OS + complexity * LP + family_size +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N250_fs)

```

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * OS + complexity * LP + family_size + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_nw, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 29933.2
||
|| Scaled residuals:
|| Min 1Q Median 3Q Max
|| -7.7366 -0.5107 -0.0056 0.4938 12.2676
||

```

```

|| Random effects:
|| Groups Name Variance Std.Dev.
|| SubjID:chlabel (Intercept) 3.109 1.763
|| SubjID (Intercept) 2.163 1.471
|| Residual 4.072 2.018
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
|| Estimate Std. Error df t value Pr(>|t|)
|| (Intercept) -0.66509 0.19682 56.99997 -3.379 0.001317 **
|| complexity1 -0.23263 0.05023 4856.00002 -4.631 3.73e-06 ***
|| OS 0.29808 0.22013 56.99997 1.354 0.181051
|| LP -0.20675 0.16003 56.99997 -1.292 0.201577
|| family_size1 0.18172 0.05014 4856.00002 3.625 0.000292 ***
|| complexity1:OS 0.39199 0.05618 4856.00002 6.978 3.40e-12 ***
|| complexity1:LP 0.35764 0.04084 4856.00002 8.758 < 2e-16 ***
|| ---
|| Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
|| (Intr) cmplx1 OS LP fmly_1 cm1:OS
|| complexity1 0.000
|| OS -0.046 0.000
|| LP 0.038 0.000 0.021
|| family_size1 0.000 0.000 0.000 0.000
|| cmplxty1:OS 0.000 -0.046 0.000 0.000 0.000
|| cmplxty1:LP 0.000 0.038 0.000 0.000 0.000 0.021
||
|| # Effect sizes for individual terms
eta_squared(m_nw_N250_fs, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----
|| complexity | 4.40e-03 | [0.00, 1.00]
|| OS | 0.03 | [0.00, 1.00]
|| LP | 0.03 | [0.00, 1.00]
|| family_size | 2.70e-03 | [0.00, 1.00]
|| complexity:OS | 9.93e-03 | [0.01, 1.00]
|| complexity:LP | 0.02 | [0.01, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
r2_nakagawa(m_nw_N250_fs)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.575
|| Marginal R2: 0.025

```

3.5.2 Means and Contrasts

```

## complexity
emmmeans(m_nw_N250_fs, ~ complexity)

|| complexity emmean SE df lower.CL upper.CL
|| Simple -0.526 0.198 58.9 -0.922 -0.130
|| Complex -0.759 0.198 58.9 -1.156 -0.363
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

## complexity x LP
emm_fs_LP <- emmeans(m_nw_N250_fs,
  ~ complexity | LP,
  at = list(LP = c(-2, -1, 0, 1, 2)))

emm_part <- summary(emm_fs_LP) |>
as.data.frame() |>
transmute(LP, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
  lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_fs_LP) |>
summary() |>
as.data.frame() |>
transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
  lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

```

```

(emm_contrast_table_fs_LP <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  ))

|| LP Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| 1 -2 Contrast: Simple - Complex 0.93 0.09 4856.0 NA NA 9.89 0.000
|| 2 -2 EMM: Complex -0.70 0.37 58.9 -1.45 0.04 NA NA
|| 3 -2 EMM: Simple 0.23 0.37 58.9 -0.52 0.97 NA NA
|| 4 -1 Contrast: Simple - Complex 0.57 0.06 4856.0 NA NA 9.05 0.000
|| 5 -1 EMM: Complex -0.73 0.25 58.9 -1.23 -0.23 NA NA
|| 6 -1 EMM: Simple -0.16 0.25 58.9 -0.66 0.34 NA NA
|| 7 0 Contrast: Simple - Complex 0.22 0.05 4856.0 NA NA 4.31 0.000
|| 8 0 EMM: Complex -0.76 0.20 58.9 -1.16 -0.36 NA NA
|| 9 0 EMM: Simple -0.54 0.20 58.9 -0.94 -0.15 NA NA
|| 10 1 Contrast: Simple - Complex -0.14 0.07 4856.0 NA NA -2.14 0.032
|| 11 1 EMM: Complex -0.79 0.26 58.9 -1.31 -0.27 NA NA
|| 12 1 EMM: Simple -0.93 0.26 58.9 -1.45 -0.41 NA NA
|| 13 2 Contrast: Simple - Complex -0.50 0.10 4856.0 NA NA -5.12 0.000
|| 14 2 EMM: Complex -0.82 0.39 58.9 -1.59 -0.05 NA NA
|| 15 2 EMM: Simple -1.32 0.39 58.9 -2.09 -0.54 NA NA

## complexity x OS
emm_fs_OS <- emmeans(m_nw_N250_fs,
  ~ complexity | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)))

emm_part <- summary(emm_fs_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_fs_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

(emm_contrast_table_fs_OS <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|> # sorts rows in the combined table, first by column "OS", then by "Effect"
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  ))

```

```

|| OS Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| 1 -1.9 Contrast: Simple - Complex 0.99 0.12 4856.0 NA NA 8.29 0.000
|| 2 -1.9 EMM: Complex -1.72 0.47 58.9 -2.67 -0.77 NA NA
|| 3 -1.9 EMM: Simple -0.72 0.47 58.9 -1.67 0.22 NA NA
|| 4 -1.0 Contrast: Simple - Complex 0.64 0.08 4856.0 NA NA 8.33 0.000
|| 5 -1.0 EMM: Complex -1.27 0.30 58.9 -1.88 -0.66 NA NA
|| 6 -1.0 EMM: Simple -0.63 0.30 58.9 -1.24 -0.02 NA NA
|| 7 0.0 Contrast: Simple - Complex 0.25 0.05 4856.0 NA NA 4.98 0.000
|| 8 0.0 EMM: Complex -0.78 0.20 58.9 -1.18 -0.38 NA NA
|| 9 0.0 EMM: Simple -0.53 0.20 58.9 -0.93 -0.13 NA NA
|| 10 1.0 Contrast: Simple - Complex -0.14 0.07 4856.0 NA NA -1.93 0.053
|| 11 1.0 EMM: Complex -0.29 0.29 58.9 -0.87 0.30 NA NA
|| 12 1.0 EMM: Simple -0.43 0.29 58.9 -1.01 0.15 NA NA
|| 13 1.9 Contrast: Simple - Complex -0.49 0.12 4856.0 NA NA -4.27 0.000
|| 14 1.9 EMM: Complex 0.16 0.46 58.9 -0.76 1.07 NA NA
|| 15 1.9 EMM: Simple -0.34 0.46 58.9 -1.25 0.58 NA NA

```

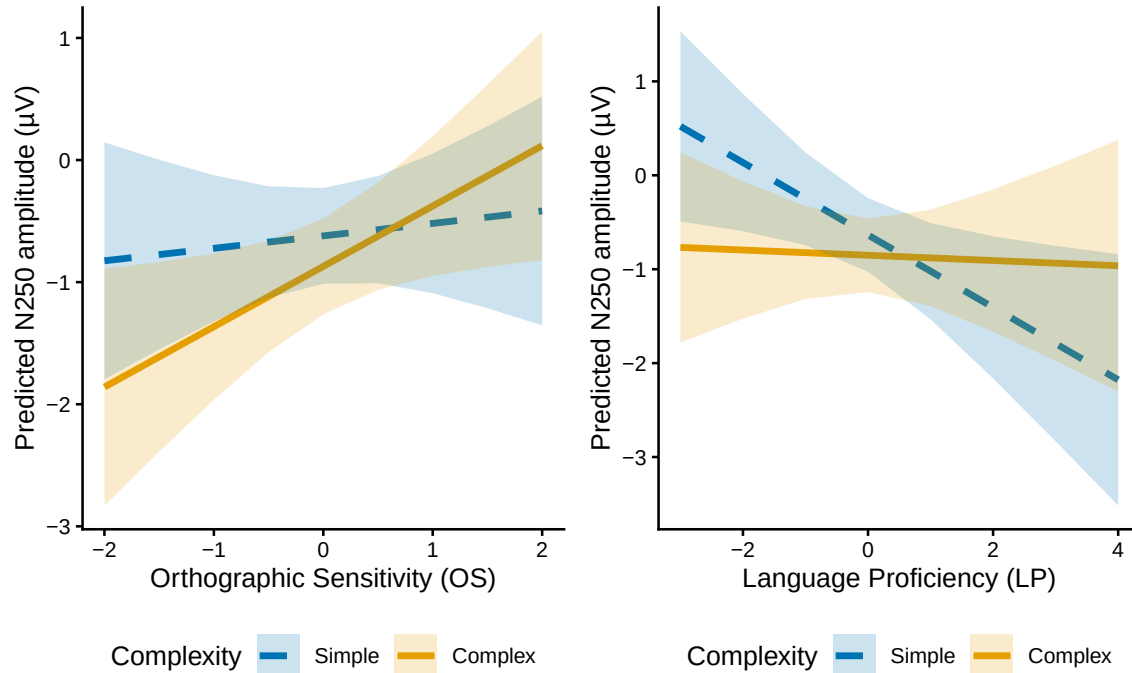
3.5.3 Plots

3.6 N250 NONWORD DATA: Combined Interaction Plots

```
row1 <- plot_grid( CMP.OS.fs.n250.nw, CMP.LP.fs.n250.nw, ncol = 2)
```

```
plot_grid(
  ggdraw() + draw_label("N250 NonWord Data", fontface = "bold"),
  row1,
  ncol = 1,
  rel_heights = c(0.08, 1)
)
```

N250 NonWord Data



4 N400

4.1 N400 WORD DATA: Base Frequency

4.1.1 Model

```
contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N400_bf <- lmer(
  value ~ base_frequency * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_bf)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 19468.7
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -11.6094  -0.3522   0.0107   0.3807  10.5439
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept) 23.249   4.822
|| SubjID       (Intercept) 18.541   4.306
|| Residual                9.157    3.026
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
```

```

|| Fixed effects:
||
||           Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)  5.617e-01  5.722e-01  5.700e+01  0.982  0.3304
|| base_frequency1  2.141e-01  1.064e-01  1.618e+03  2.012  0.0444 *
|| LP           4.677e-02  4.652e-01  5.700e+01  0.101  0.9203
|| OS           1.290e+00  6.399e-01  5.700e+01  2.016  0.0485 *
|| base_frequency1:LP  3.444e-01  8.659e-02  1.618e+03  3.977  7.29e-05 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||           (Intr) bs_fr1 LP      OS
|| bas_frncyl  0.000
|| LP          0.038  0.000
|| OS          -0.046  0.000  0.021
|| bs_frqn1:LP  0.000  0.039  0.000  0.000

```

Effect sizes for individual terms

```

eta_squared(m_w_N400_bf, partial = TRUE)

```

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----
|| base_frequency |      2.50e-03 | [0.00, 1.00]
|| LP             |      1.77e-04 | [0.00, 1.00]
|| OS             |       0.07 | [0.00, 1.00]
|| base_frequency:LP |     9.68e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

Model R²

```

r2_nakagawa(m_w_N400_bf)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.825
|| Marginal R2: 0.026

```

4.1.2 Means and Contrasts

```

emmeans(m_w_N400_bf, ~ base_frequency)

```

```

|| base_frequency emmean SE df lower.CL upper.CL
|| Low           0.515 0.574 58 -0.634 1.66
|| High          0.712 0.574 58 -0.436 1.86
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

```

```

emm_bf_LP <- emmeans(m_w_N400_bf,
  ~ base_frequency | LP,
  at = list(LP = c(-2, -1, 0, 1, 2)))

```

```

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

```

```

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

```

```

(emm_contrast_table_LP_N400_words <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  ))

```

```

|| LP           Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| 1 -2 Contrast: Low - High  0.47 0.20 1618      NA      NA  2.38  0.018
|| 2 -2           EMM: High   0.28 1.08  58 -1.87  2.44      NA      NA
|| 3 -2           EMM: Low    0.76 1.08  58 -1.40  2.92      NA      NA

```

```

|| 4 -1 Contrast: Low - High      0.13 0.13 1618      NA      NA      0.97 0.333
|| 5 -1      EMM: High           0.50 0.73 58      -0.95      1.96      NA      NA
|| 6 -1      EMM: Low            0.63 0.73 58      -0.82      2.09      NA      NA
|| 7 0 Contrast: Low - High     -0.21 0.11 1618      NA      NA     -2.01 0.044
|| 8 0      EMM: High            0.72 0.57 58      -0.43      1.87      NA      NA
|| 9 0      EMM: Low             0.51 0.57 58      -0.64      1.66      NA      NA
|| 10 1 Contrast: Low - High    -0.56 0.14 1618      NA      NA     -3.99 0.000
|| 11 1      EMM: High            0.94 0.75 58      -0.57      2.45      NA      NA
|| 12 1      EMM: Low             0.38 0.75 58     -1.13      1.89      NA      NA
|| 13 2 Contrast: Low - High    -0.90 0.21 1618      NA      NA     -4.37 0.000
|| 14 2      EMM: High            1.16 1.12 58     -1.07      3.39      NA      NA
|| 15 2      EMM: Low             0.26 1.12 58     -1.98      2.49      NA      NA
# kable(emm_contrast_table, digits = 3)

```

4.1.3 Plots

4.2 N400 WORD DATA: Family Size

4.2.1 Model

```

contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N400_fs <- lmer(
  value ~ family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_fs)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15561.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -10.9307  -0.3874   0.0081   0.4071   6.2048
||
|| Random effects:
|| Groups          Name      Variance Std.Dev.
|| SubjID:chlabel (Intercept) 5.238    2.289
|| SubjID         (Intercept) 4.575    2.139
|| Residual                3.304    1.818
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)  2.618e-01  2.842e-01  5.700e+01  0.921  0.3608
|| family_size1  4.522e-01  6.392e-02  1.618e+03  7.074  2.23e-12 ***
|| LP           1.231e-03  2.311e-01  5.700e+01  0.005  0.9958
|| OS           6.771e-01  3.179e-01  5.700e+01  2.130  0.0375 *
|| family_size1:LP -7.114e-02  5.201e-02  1.618e+03 -1.368  0.1716
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) fmly_1 LP      OS
|| family_siz1  0.000
|| LP           0.038  0.000
|| OS          -0.046  0.000  0.021
|| fmly_sz1:LP  0.000  0.039  0.000  0.000
# Effect sizes for individual terms
eta_squared(m_w_N400_fs, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----
|| family_size    |          0.03 | [0.02, 1.00]
|| LP             |       4.98e-07 | [0.00, 1.00]
|| OS             |          0.07 | [0.00, 1.00]
|| family_size:LP |       1.15e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```



```
# Model R²
r2_nakagawa(m_w_N400_fs)
```

```
|| # R2 for Mixed Models
||
|| Conditional R2: 0.756
|| Marginal R2: 0.031
```

4.2.2 Means

```
emmeans(m_w_N400_fs, ~ family_size)
```

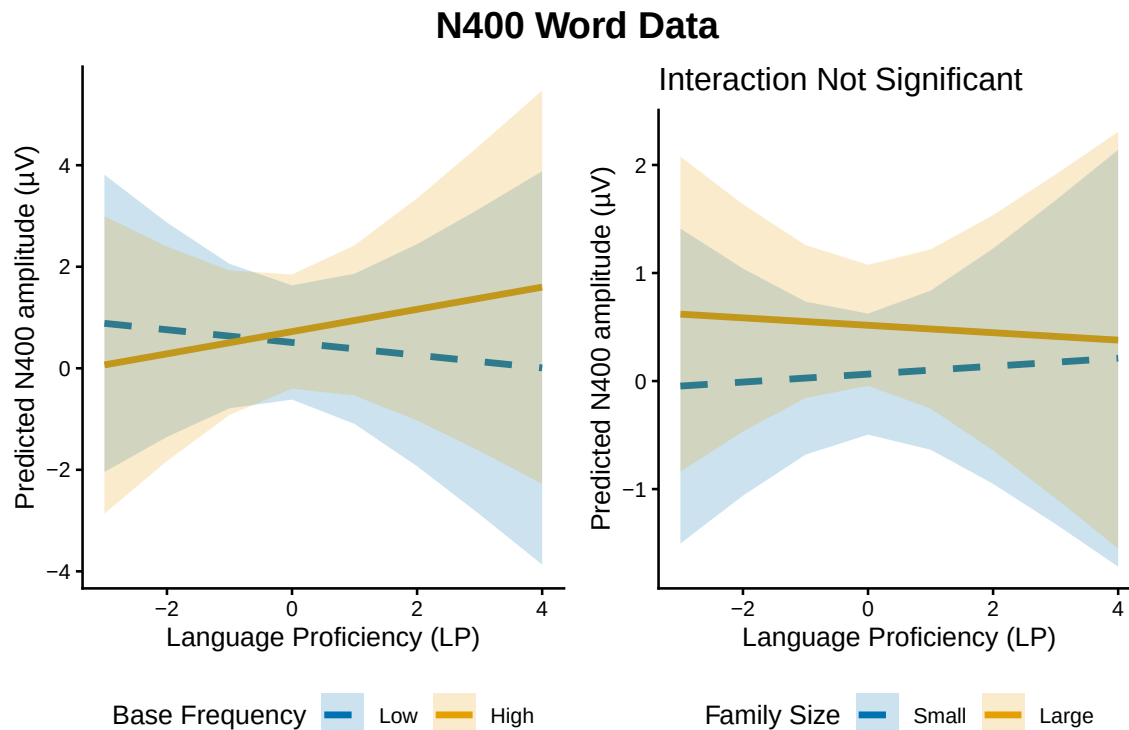
```
|| family_size emmean SE df lower.CL upper.CL
|| Small 0.0623 0.286 58.5 -0.5091 0.634
|| Large 0.5179 0.286 58.5 -0.0535 1.089
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
```

4.2.3 Plots (Interaction Not significant)

4.3 N400 WORD DATA: Combined Interaction Plots

```
row1 <- plot_grid(BFxLP_n400,FSxLP_n400,ncol = 2)

plot_grid(
  ggdraw() + draw_label("N400 Word Data", fontface = "bold"),
  row1,
  ncol = 1,
  rel_heights = c(0.08, 1)
)
```



Does the slopes of the lines in the significant interactions, significantly differ from zero?

```
test(entrends(m_w_N250_bf, ~ base_frequency, var = "OS"))
```

```
|| base_frequency OS.trend SE df t.ratio p.value
|| Low 0.762 0.523 58.3 1.456 0.1507
|| High 0.307 0.523 58.3 0.586 0.5600
||
|| Degrees-of-freedom method: kenward-roger
```

```
test(entrends(m_w_N400_bf, ~ base_frequency, var = "LP"))
```

```

|| base_frequency LP.trend SE df t.ratio p.value
|| Low -0.125 0.467 58 -0.268 0.7893
|| High 0.219 0.467 58 0.469 0.6411
||
|| Degrees-of-freedom method: kenward-roger

```

4.3.1 N400 NONWORD DATA: Base Frequency (NOT USED IN WRITE-UP)

4.3.2 Model

```

contrasts(erp_bf_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_bf_nw$base_frequency) <- c(-0.5, 0.5)

m_nw_N400_bf <- lmer(
  value ~ complexity * base_frequency * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N400_bf)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * base_frequency * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_nw, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 44167.9
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -11.9475  -0.3895   0.0032   0.4332  20.9414
||
|| Random effects:
|| Groups             Name             Variance Std.Dev.
|| SubjID:chlabel (Intercept)  23.71      4.869
|| SubjID           (Intercept)  23.96      4.895
|| Residual                    38.00      6.164
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -0.84542    0.64909   56.99999   -1.302  0.19799
|| complexity1     -0.44211    0.15327  4854.00000   -2.884  0.00394 **
|| base_frequency1 -0.04368    0.15327  4854.00000   -0.285  0.77566
|| LP              0.23770    0.52775   56.99999    0.450  0.65413
|| OS              0.74117    0.72597   56.99999    1.021  0.31159
|| complexity1:base_frequency1 -0.05967    0.30655  4854.00000   -0.195  0.84566
|| complexity1:LP    0.20562    0.12473  4854.00000    1.649  0.09930
|| base_frequency1:LP 0.11261    0.12473  4854.00000    0.903  0.36663
|| complexity1:base_frequency1:LP -0.17117    0.24945  4854.00000   -0.686  0.49264
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) cmplx1 bs_fr1 LP      OS      cm1:_1 cm1:LP b_1:LP
|| complexity1  0.000
|| bas_frqn1    0.000 0.000
|| LP           0.038 0.000 0.000
|| OS           -0.046 0.000 0.000 0.021
|| cmplxty1:_1  0.000 0.000 0.000 0.000 0.000
|| cmplxty1:LP  0.000 0.039 0.000 0.000 0.000 0.000
|| bs_frqn1:LP  0.000 0.000 0.039 0.000 0.000 0.000 0.000
|| cmpl1:_1:LP  0.000 0.000 0.000 0.000 0.000 0.039 0.000 0.000

# Effect sizes for individual terms
eta_squared(m_nw_N400_bf, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----
|| complexity | 1.71e-03 | [0.00, 1.00]
|| base_frequency | 1.67e-05 | [0.00, 1.00]
|| LP | 3.55e-03 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]
|| complexity:base_frequency | 7.81e-06 | [0.00, 1.00]
|| complexity:LP | 5.60e-04 | [0.00, 1.00]
|| base_frequency:LP | 1.68e-04 | [0.00, 1.00]
|| complexity:base_frequency:LP | 9.70e-05 | [0.00, 1.00]

```

```

||
|| - One-sided CIs: upper bound fixed at [1.00].
# Model R²
r2_nakagawa(m_nw_N400_bf)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.559
|| Marginal R2: 0.007

```

4.3.3 N400 NONWORD DATA: Family Size

```

contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)

m_nw_N400_fs <- lmer(
  value ~ complexity * family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N400_fs)

```

Model

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_nw, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 32979.6
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -13.9873  -0.4837  -0.0118   0.4655  14.1595
||
|| Random effects:
|| Groups          Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  6.389      2.528
|| SubjID          (Intercept)  5.568      2.360
|| Residual                6.114      2.473
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)   -0.45163    0.31311   57.00002  -1.442 0.154666
|| complexity1   -0.45956    0.06148 4853.99999  -7.475 9.11e-14 ***
|| family_size1    0.08743    0.06148 4853.99999   1.422 0.155086
|| LP              0.01139    0.25458   57.00002   0.045 0.964463
|| OS              0.38123    0.35019   57.00002   1.089 0.280904
|| complexity1:family_size1 -0.44008    0.12296 4853.99999  -3.579 0.000348 ***
|| complexity1:LP    0.25043    0.05003 4853.99999   5.006 5.77e-07 ***
|| family_size1:LP    0.12120    0.05003 4853.99999   2.423 0.015450 *
|| complexity1:family_size1:LP 0.22717    0.10006 4853.99999   2.270 0.023230 *
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) cmplx1 fmly_1 LP      OS      cm1:_1 cm1:LP f_1:LP
|| complexity1  0.000
|| family_size1 0.000 0.000
|| LP           0.038 0.000 0.000
|| OS          -0.046 0.000 0.000 0.021
|| cmplxty1:_1 0.000 0.000 0.000 0.000 0.000
|| cmplxty1:LP 0.000 0.000 0.039 0.000 0.000 0.000
|| fmly_sz1:LP 0.000 0.000 0.039 0.000 0.000 0.000 0.000
|| cmpl1:_1:LP 0.000 0.000 0.000 0.000 0.000 0.039 0.000 0.000
# Effect sizes for individual terms
eta_squared(m_nw_N400_fs, partial = TRUE)

```

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| complexity | 0.01 | [0.01, 1.00]
|| family_size | 4.16e-04 | [0.00, 1.00]
|| LP | 3.51e-05 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]

```

```

|| complexity:family_size | 2.63e-03 | [0.00, 1.00]
|| complexity:LP | 5.14e-03 | [0.00, 1.00]
|| family_size:LP | 1.21e-03 | [0.00, 1.00]
|| complexity:family_size:LP | 1.06e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# Model R²
r2_nakagawa(m_nw_N400_fs)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.666
|| Marginal R2: 0.012

```

```

# emm_3way <- emmeans(
#   m_nw_N400_fs,
#   ~ complexity * family_size | LP,
#   at = list(LP = c(-1, 0, 1))
# )

```

```

emm_3way <- emmeans(
  m_nw_N400_fs,
  ~ complexity * family_size | LP,
  at = list(LP = c(-2,-1, 0, 1, 2))
)

```

```

compact_3way_table(
  emm_3way,
  by_var = LP,
  condition_prefix = "LP",
  factor1 = complexity,
  factor2 = family_size
)

```

Means and Contrasts: Complexity × Family Size × LP interaction

	Comparison	Condition	Estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
1	Complex Large	LP = -1	-0.99	0.40	60.4	-1.79	-0.18	NA	NA
2	Complex Small	LP = -1	-0.62	0.40	60.4	-1.42	0.18	NA	NA
3	Simple Large	LP = -1	0.06	0.40	60.4	-0.74	0.86	NA	NA
4	Simple Small	LP = -1	-0.24	0.40	60.4	-1.04	0.56	NA	NA
5	Int: Simple - Complex across	LP = -1	-0.67	0.16	4854.0	NA	NA	-4.29	0.000
6	Small - Large	LP = -1 complexity = Complex	0.37	0.11	4854.0	NA	NA	3.34	0.001
7	Small - Large	LP = -1 complexity = Simple	-0.30	0.11	4854.0	NA	NA	-2.73	0.006
8	Simple - Complex	LP = -1 family_size = Large	1.04	0.11	4854.0	NA	NA	9.49	0.000
9	Simple - Complex	LP = -1 family_size = Small	0.38	0.11	4854.0	NA	NA	3.42	0.001
10	Complex Large	LP = -2	-1.24	0.60	60.3	-2.43	-0.05	NA	NA
11	Complex Small	LP = -2	-0.64	0.60	60.3	-1.83	0.55	NA	NA
12	Simple Large	LP = -2	0.17	0.60	60.3	-1.02	1.36	NA	NA
13	Simple Small	LP = -2	-0.12	0.60	60.3	-1.32	1.07	NA	NA
14	Int: Simple - Complex across	LP = -2	-0.89	0.23	4854.0	NA	NA	-3.88	0.000
15	Small - Large	LP = -2 complexity = Complex	0.60	0.16	4854.0	NA	NA	3.69	0.000
16	Small - Large	LP = -2 complexity = Simple	-0.29	0.16	4854.0	NA	NA	-1.79	0.073
17	Simple - Complex	LP = -2 family_size = Large	1.41	0.16	4854.0	NA	NA	8.63	0.000
18	Simple - Complex	LP = -2 family_size = Small	0.51	0.16	4854.0	NA	NA	3.15	0.002
19	Complex Large	LP = 0	-0.73	0.32	60.4	-1.37	-0.10	NA	NA
20	Complex Small	LP = 0	-0.60	0.32	60.4	-1.23	0.04	NA	NA
21	Simple Large	LP = 0	-0.05	0.32	60.4	-0.69	0.58	NA	NA
22	Simple Small	LP = 0	-0.36	0.32	60.4	-0.99	0.27	NA	NA
23	Int: Simple - Complex across	LP = 0	-0.44	0.12	4854.0	NA	NA	-3.58	0.000
24	Small - Large	LP = 0 complexity = Complex	0.13	0.09	4854.0	NA	NA	1.53	0.127
25	Small - Large	LP = 0 complexity = Simple	-0.31	0.09	4854.0	NA	NA	-3.54	0.000
26	Simple - Complex	LP = 0 family_size = Large	0.68	0.09	4854.0	NA	NA	7.82	0.000
27	Simple - Complex	LP = 0 family_size = Small	0.24	0.09	4854.0	NA	NA	2.75	0.006
28	Complex Large	LP = 1	-0.48	0.42	60.3	-1.31	0.36	NA	NA
29	Complex Small	LP = 1	-0.58	0.42	60.3	-1.41	0.25	NA	NA
30	Simple Large	LP = 1	-0.16	0.42	60.3	-1.00	0.67	NA	NA
31	Simple Small	LP = 1	-0.48	0.42	60.3	-1.31	0.36	NA	NA
32	Int: Simple - Complex across	LP = 1	-0.21	0.16	4854.0	NA	NA	-1.32	0.188
33	Small - Large	LP = 1 complexity = Complex	-0.10	0.11	4854.0	NA	NA	-0.89	0.371
34	Small - Large	LP = 1 complexity = Simple	-0.32	0.11	4854.0	NA	NA	-2.76	0.006
35	Simple - Complex	LP = 1 family_size = Large	0.32	0.11	4854.0	NA	NA	2.76	0.006
36	Simple - Complex	LP = 1 family_size = Small	0.10	0.11	4854.0	NA	NA	0.90	0.369
37	Complex Large	LP = 2	-0.22	0.62	60.3	-1.46	1.01	NA	NA
38	Complex Small	LP = 2	-0.56	0.62	60.3	-1.79	0.67	NA	NA
39	Simple Large	LP = 2	-0.27	0.62	60.3	-1.51	0.96	NA	NA
40	Simple Small	LP = 2	-0.59	0.62	60.3	-1.83	0.64	NA	NA
41	Int: Simple - Complex across	LP = 2	0.01	0.24	4854.0	NA	NA	0.06	0.952
42	Small - Large	LP = 2 complexity = Complex	-0.34	0.17	4854.0	NA	NA	-1.99	0.046

```

|| 43          Small - Large   LP = 2 | complexity = Simple   -0.32 0.17 4854.0      NA      NA   -1.91   0.056
|| 44          Simple - Complex LP = 2 | family_size = Large  -0.05 0.17 4854.0      NA      NA   -0.29   0.774
|| 45          Simple - Complex LP = 2 | family_size = Small  -0.03 0.17 4854.0      NA      NA   -0.20   0.840

```

```

emm_complexity_FS_N400 <- emmeans(
  m_nw_N400_fs,
  ~ complexity | family_size)

compact_emm_table(
  emm_complexity_FS_N400,
  by_var = family_size
)

```

Complexity × Family Size interaction averaged across LP

```

|| # A tibble: 6 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex      Large      -0.74 0.32   60.4  -1.38  -0.11    NA     NA
|| 2 Simple      Large      -0.05 0.32   60.4  -0.68  0.59    NA     NA
|| 3 Simple - Complex Large      0.7   0.09 4854    NA     NA      8.02    0
|| 4 Complex      Small      -0.6   0.32   60.4  -1.23  0.03    NA     NA
|| 5 Simple      Small      -0.35 0.32   60.4  -0.99  0.28    NA     NA
|| 6 Simple - Complex Small      0.25 0.09 4854    NA     NA      2.83   0.005

```

```

emm_complexity_CMP_N400 <- emmeans(
  m_nw_N400_fs,
  ~ family_size | complexity)

compact_emm_table(
  emm_complexity_CMP_N400,
  by_var = complexity
)

```

```

|| # A tibble: 6 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Large      Complex      -0.74 0.32   60.4  -1.38  -0.11    NA     NA
|| 2 Small      Complex      -0.6   0.32   60.4  -1.23  0.03    NA     NA
|| 3 Small - Large Complex      0.14 0.09 4854    NA     NA      1.66   0.098
|| 4 Large      Simple      -0.05 0.32   60.4  -0.68  0.59    NA     NA
|| 5 Small      Simple      -0.35 0.32   60.4  -0.99  0.28    NA     NA
|| 6 Small - Large Simple      -0.31 0.09 4854    NA     NA     -3.53    0

```

```

emm_complexity_LP_N400 <- emmeans(
  m_nw_N400_fs,
  ~ complexity | LP,
  at = list(LP = c(-2, -1, 0, 1, 2)))

compact_emm_table(
  emm_complexity_LP_N400,
  by_var = LP,
  condition_prefix = "LP"
)

```

Complexity × LP interaction averaged over Family Size:

```

|| # A tibble: 15 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex      LP = -1      -0.8   0.4   58.1  -1.6  -0.01    NA     NA
|| 2 Simple      LP = -1     -0.09 0.4   58.1  -0.89  0.7     NA     NA
|| 3 Simple - Complex LP = -1      0.71 0.08 4854    NA     NA      9.13    0
|| 4 Complex      LP = -2     -0.94 0.59 58.1  -2.12  0.24    NA     NA
|| 5 Simple      LP = -2      0.02 0.59 58.1  -1.16  1.2     NA     NA
|| 6 Simple - Complex LP = -2      0.96 0.12 4854    NA     NA      8.33    0
|| 7 Complex      LP = 0      -0.67 0.31 58.1  -1.29 -0.04    NA     NA
|| 8 Simple      LP = 0     -0.21 0.31 58.1  -0.84  0.42    NA     NA
|| 9 Simple - Complex LP = 0      0.46 0.06 4854    NA     NA      7.47    0
|| 10 Complex     LP = 1     -0.53 0.41 58.1  -1.36  0.3     NA     NA
|| 11 Simple     LP = 1     -0.32 0.41 58.1  -1.15  0.51    NA     NA
|| 12 Simple - Complex LP = 1      0.21 0.08 4854    NA     NA      2.59   0.01
|| 13 Complex     LP = 2     -0.39 0.61 58.1  -1.61  0.83    NA     NA
|| 14 Simple     LP = 2     -0.43 0.61 58.1  -1.66  0.79    NA     NA
|| 15 Simple - Complex LP = 2     -0.04 0.12 4854    NA     NA     -0.35   0.73

```

```

emm_FS_LP_N400 <- emmeans(

```

```

m_nw_N400_fs,
~ family_size | LP,
at = list(LP = c(-2, -1, 0, 1, 2))
)

compact_emm_table(
  emm_FS_LP_N400,
  by_var = LP,
  condition_prefix = "LP"
)

```

FS × LP interaction averaged over Complexity:

```

|| # A tibble: 15 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Large         LP = -1      -0.46  0.4   58.1   -1.26  0.33   NA     NA
|| 2 Small         LP = -1      -0.43  0.4   58.1   -1.23  0.37   NA     NA
|| 3 Small - Large LP = -1        0.03  0.08  4854    NA     NA     0.43  0.664
|| 4 Large         LP = -2      -0.54  0.59  58.1   -1.72  0.64   NA     NA
|| 5 Small         LP = -2      -0.38  0.59  58.1   -1.56  0.8     NA     NA
|| 6 Small - Large LP = -2        0.15  0.12  4854    NA     NA     1.34  0.179
|| 7 Large         LP = 0       -0.39  0.31  58.1   -1.02  0.24   NA     NA
|| 8 Small         LP = 0       -0.48  0.31  58.1   -1.11  0.15   NA     NA
|| 9 Small - Large LP = 0       -0.09  0.06  4854    NA     NA    -1.42  0.155
|| 10 Large        LP = 1      -0.32  0.41  58.1   -1.15  0.51   NA     NA
|| 11 Small        LP = 1      -0.53  0.41  58.1   -1.36  0.3     NA     NA
|| 12 Small - Large LP = 1      -0.21  0.08  4854    NA     NA    -2.58  0.01
|| 13 Large        LP = 2      -0.25  0.61  58.1   -1.47  0.97   NA     NA
|| 14 Small        LP = 2      -0.58  0.61  58.1   -1.8   0.64   NA     NA
|| 15 Small - Large LP = 2      -0.33  0.12  4854    NA     NA    -2.76  0.006

```

```

emm_options(lmer.df = "satterthwaite")
emmmeans(m_nw_N400_fs, pairwise ~ complexity | family_size)

```

complexity effects at each family size level averaged over LP

```

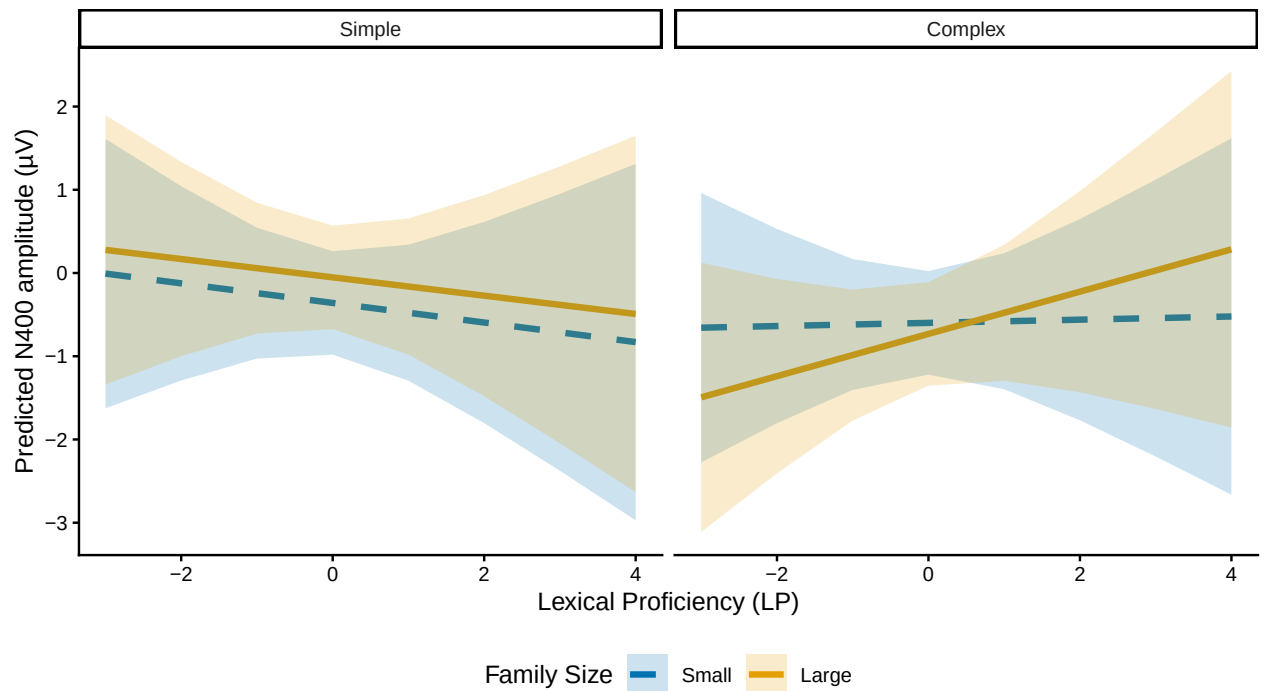
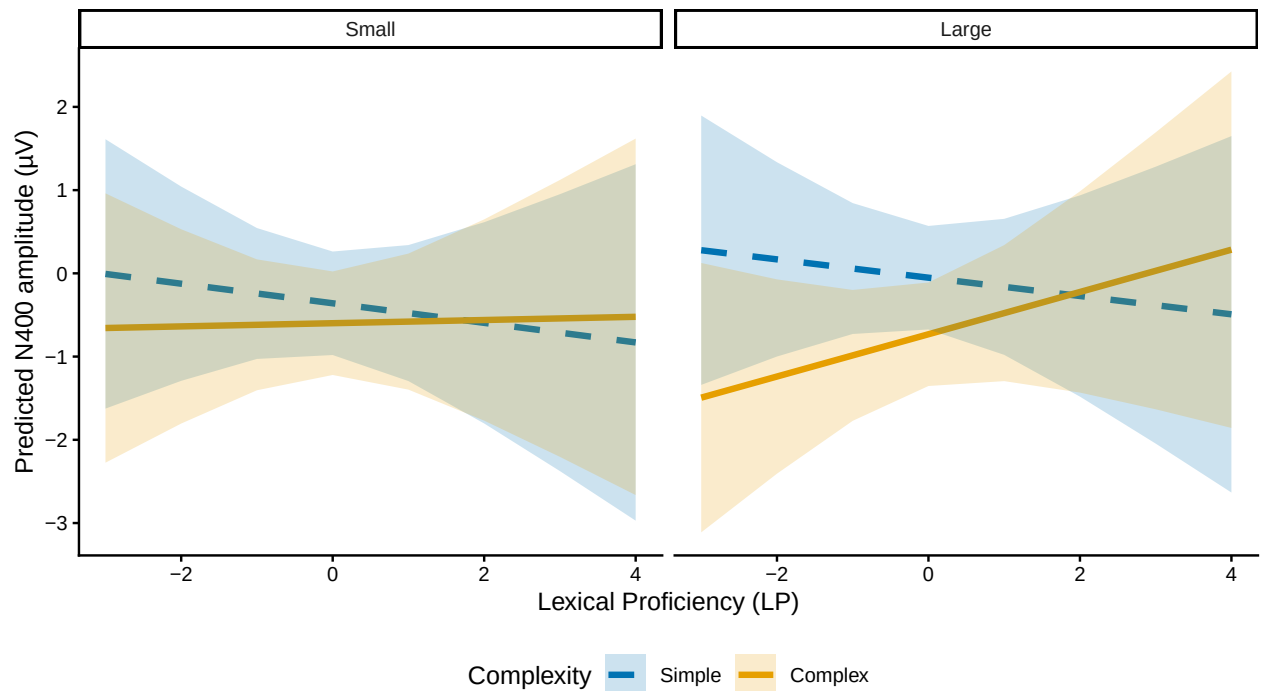
|| $emmmeans
|| family_size = Small:
||   complexity emmean      SE      df lower.CL upper.CL
|| Simple      -0.3540 0.317 60.4   -0.988  0.280
|| Complex     -0.6001 0.317 60.4   -1.234  0.034
||
|| family_size = Large:
||   complexity emmean      SE      df lower.CL upper.CL
|| Simple      -0.0469 0.317 60.4   -0.681  0.587
|| Complex     -0.7440 0.317 60.4   -1.378  -0.110
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
||
|| $contrasts
|| family_size = Small:
||   contrast      estimate      SE      df t.ratio p.value
|| Simple - Complex  0.246 0.0869 4854    2.833  0.0046
||
|| family_size = Large:
||   contrast      estimate      SE      df t.ratio p.value
|| Simple - Complex  0.697 0.0869 4854    8.025 <.0001
||
|| Degrees-of-freedom method: kenward-roger

```

```

plot_grid(nw.n400.3way.A, nw.n400.3way.B, ncol = 1)

```



Plots

```
library(emmeans)
library(dplyr)
library(ggplot2)

# Create a sequence of LP values across the observed range
LP_seq <- with(
  subset(erp_bf_nw, time_window == "N400"),
  seq(min(LP, na.rm = TRUE),
      max(LP, na.rm = TRUE),
      length.out = 100)
)

# Estimate complexity means at each LP value, separately by family_size
emm_complexity_by_LP_FS <- emmeans(
```

```

m_nw_N400_fs,
~ complexity | family_size * LP,
at = list(LP = LP_seq)
)

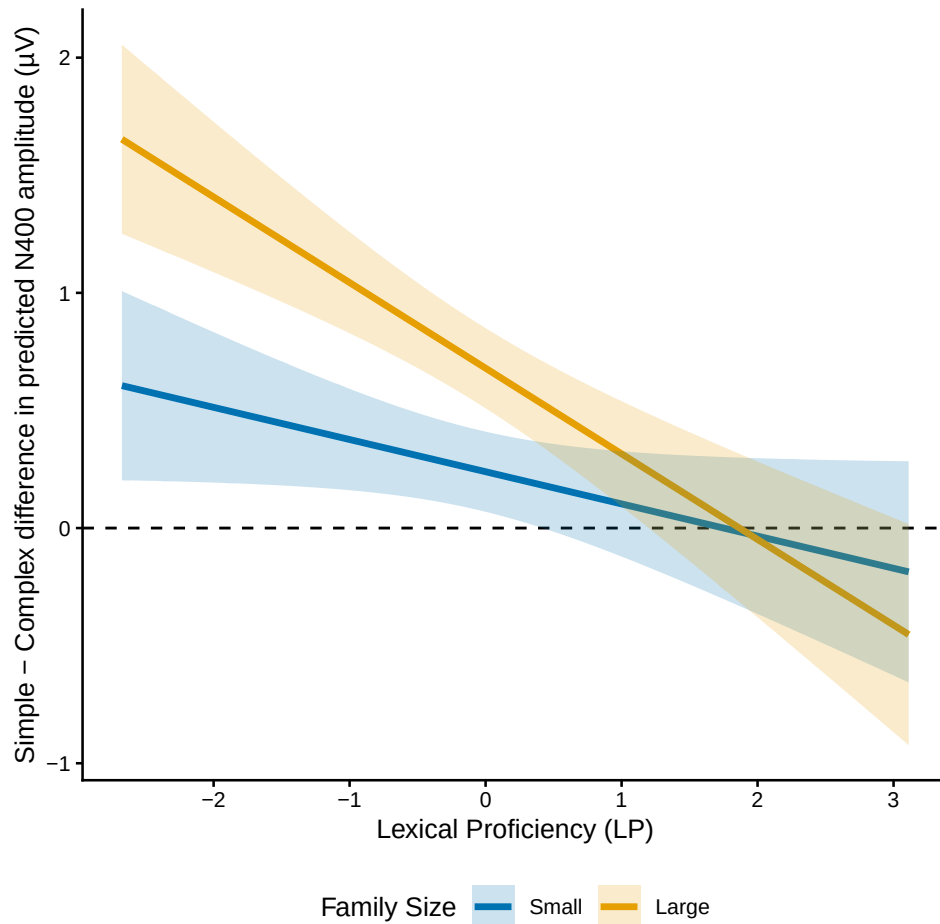
# Compute Simple - Complex at each LP value and family_size level
complexity_diff_by_LP_FS <- pairs(emm_complexity_by_LP_FS) |>
  summary(infer = TRUE) |>
  as.data.frame()

head(complexity_diff_by_LP_FS)

|| contrast          family_size      LP estimate      SE df lower.CL upper.CL t.ratio p.value
|| Simple - Complex Small      -2.674523 0.6055103 0.2051206 4854 0.2033810 1.0076395 2.952 0.0032
|| Simple - Complex Large      -2.674523 1.6531584 0.2051206 4854 1.2510292 2.0552877 8.059 <.0001
|| Simple - Complex Small      -2.616080 0.5975128 0.2013825 4854 0.2027119 0.9923138 2.967 0.0030
|| Simple - Complex Large      -2.616080 1.6318848 0.2013825 4854 1.2370839 2.0266857 8.103 <.0001
|| Simple - Complex Small      -2.557638 0.5895154 0.1976602 4854 0.2020118 0.9770190 2.982 0.0029
|| Simple - Complex Large      -2.557638 1.6106112 0.1976602 4854 1.2231076 1.9981147 8.148 <.0001
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

ggplot(
  complexity_diff_by_LP_FS,
  aes(x = LP,
      y = estimate,
      colour = family_size,
      fill = family_size)
) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             linewidth = 0.5) +
  geom_line(linewidth = 1.2) +
  geom_ribbon(aes(ymin = lower.CL,
                 ymax = upper.CL),
             alpha = 0.20,
             colour = NA) +
  scale_colour_manual(values = my_palette_2) +
  scale_fill_manual(values = my_palette_2) +
  labs(
    x = "Lexical Proficiency (LP)",
    y = "Simple - Complex difference in predicted N400 amplitude (µV)",
    colour = "Family Size",
    fill = "Family Size"
  ) +
  my_theme +
  my_guides

```

5 EXPLORATORY ANALYSES

```
m_rt_nw_os <- lmer(
  logRT ~ complexity * OS + complexity * LP + zLogBF + zLogFS +
  (1 + complexity | SubjID) + (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
emm_options(lmer.df = "satterthwaite")
emmeans(m_rt_nw_os, pairwise ~ complexity | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)))
```

```
|| $emmeans
|| OS = -1.9:
|| complexity emmean      SE    df lower.CL upper.CL
|| Simple      6.605 0.0396 65.23    6.525    6.684
|| Complex      6.658 0.0388 65.37    6.580    6.735
||
|| OS = -1.0:
|| complexity emmean      SE    df lower.CL upper.CL
|| Simple      6.569 0.0254 66.90    6.519    6.620
|| Complex      6.621 0.0249 67.20    6.571    6.670
||
|| OS = 0.0:
|| complexity emmean      SE    df lower.CL upper.CL
|| Simple      6.530 0.0171 69.97    6.496    6.564
|| Complex      6.579 0.0168 70.26    6.546    6.613
||
|| OS = 1.0:
|| complexity emmean      SE    df lower.CL upper.CL
|| Simple      6.491 0.0256 66.29    6.439    6.542
|| Complex      6.538 0.0251 66.14    6.488    6.588
```

```

||
|| OS = 1.9:
|| complexity emmean SE df lower.CL upper.CL
|| Simple 6.455 0.0400 64.80 6.375 6.535
|| Complex 6.500 0.0391 64.60 6.422 6.578
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
||
|| $contrasts
|| OS = -1.9:
|| contrast estimate SE df t.ratio p.value
|| Simple - Complex -0.0532 0.01280 65.0 -4.167 0.0001
||
|| OS = -1.0:
|| contrast estimate SE df t.ratio p.value
|| Simple - Complex -0.0513 0.00816 66.6 -6.284 <.0001
||
|| OS = 0.0:
|| contrast estimate SE df t.ratio p.value
|| Simple - Complex -0.0491 0.00528 62.3 -9.312 <.0001
||
|| OS = 1.0:
|| contrast estimate SE df t.ratio p.value
|| Simple - Complex -0.0470 0.00782 56.7 -6.009 <.0001
||
|| OS = 1.9:
|| contrast estimate SE df t.ratio p.value
|| Simple - Complex -0.0451 0.01240 57.5 -3.646 0.0006
||
|| Degrees-of-freedom method: kenward-roger
emmeans(m_rt_nw_os, pairwise ~ complexity | LP,
  at = list(LP = c(-2, -1, 0, 1, 2)),
  tran = "log")

|| $emmeans
|| LP = -2:
|| complexity emmean SE df lower.CL upper.CL
|| Simple 6.534 0.0302 66.87 6.474 6.594
|| Complex 6.565 0.0297 67.83 6.505 6.624
||
|| LP = -1:
|| complexity emmean SE df lower.CL upper.CL
|| Simple 6.531 0.0211 68.82 6.489 6.573
|| Complex 6.570 0.0207 69.84 6.529 6.612
||
|| LP = 0:
|| complexity emmean SE df lower.CL upper.CL
|| Simple 6.529 0.0170 70.16 6.495 6.563
|| Complex 6.576 0.0167 70.58 6.543 6.610
||
|| LP = 1:
|| complexity emmean SE df lower.CL upper.CL
|| Simple 6.526 0.0212 67.44 6.484 6.568
|| Complex 6.582 0.0207 67.23 6.541 6.624
||
|| LP = 2:
|| complexity emmean SE df lower.CL upper.CL
|| Simple 6.523 0.0303 65.60 6.463 6.584
|| Complex 6.588 0.0297 65.39 6.529 6.648
||
|| Degrees-of-freedom method: kenward-roger
|| Results are given on the log (not the response) scale.
|| Confidence level used: 0.95
||
|| $contrasts
|| LP = -2:
|| contrast estimate SE df t.ratio p.value
|| Simple - Complex -0.0304 0.01020 73.7 -2.970 0.0040
||
|| LP = -1:
|| contrast estimate SE df t.ratio p.value
|| Simple - Complex -0.0391 0.00704 73.6 -5.550 <.0001
||
|| LP = 0:
|| contrast estimate SE df t.ratio p.value
|| Simple - Complex -0.0477 0.00530 64.0 -9.004 <.0001
||
|| LP = 1:
|| contrast estimate SE df t.ratio p.value

```

```

|| Simple - Complex -0.0564 0.00637 55.7 -8.858 <.0001
||
|| LP = 2:
|| contrast      estimate      SE   df t.ratio p.value
|| Simple - Complex -0.0650 0.00932 56.7 -6.980 <.0001
||
|| Degrees-of-freedom method: kenward-roger
|| Results are given on the log (not the response) scale.

```